

Kevin Zhu

Software Engineer

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Work Experience

Software Engineer	Millennium Management	July 2024 - Present
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Self-Service Portal

- Engineered a horizontally scalable ASP.NET Core service — Redis-backed shared state, Kubernetes-based replication, circuit-breaker/idempotency controls — instrumented with Prometheus/OpenTelemetry and Datadog tracing for production observability
- Built a production self-service platform unifying Microsoft Entra, PingFederate, Keyfactor PKI, Azure Key Vault, and ServiceNow, governing 9,140 applications and 1,105 certificates across 1,044 distinct owners
- Containerized the service with Docker and built a GitOps release pipeline — GitHub Actions CI building and publishing images to Artifactory, with ArgoCD deploying new versions on release

Information Security API

- Built and scaled InfosecAPI, the automation backbone behind ServiceNow-driven software, PKI certificate, and cloud-account requests, exposing a full CRUD plane over Azure applications
- Automated OAuth/SCIM provisioning across 350+ Snowflake, Datadog, and Sentry integrations through an idempotent endpoint with automatic retries and failure alerting
- Integrated Keycard for agentic identity management and built an MCP server for InfosecAPI, enabling authenticated LLM/agent access to internal identity automation

Software Engineer Intern	Millennium Management	June 2023 - August 2023
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- Built an automated Keyfactor certificate-renewal workflow covering ~2,000 applications, eliminating recurring certificate-expiration outages
- Automated Azure AD application-expiration monitoring and owner alerting, replacing manual tracking
- Built a self-service privileged-access recertification dashboard, centralizing access reviews
- Automated security incident report generation, cutting manual reporting time in incident response

Education

B.S. Electrical Engineering and Computer Science	University of California, Berkeley	August 2020 - May 2024
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Projects

Semiconductor Supply-Chain Research Pipeline & Local RAG Search

- Built and scheduled a resilient multi-source data pipeline (SEC EDGAR XBRL, Yahoo, Alpha Vantage, FMP) maintaining full financial statements for 72 supply-chain companies — ~184K financial facts, 37K daily price bars — with retry/backoff, stale-lock detection, and unattended orchestration
- Root-caused a ticker-tagging bug where 75.7% of 3,973 tagged mentions were false positives; rebuilt the matcher with cashtag/capitalization-aware evidence tiers, cutting mismatched tags to 1,832 while recovering 11 missed tickers, verified by a 30-test regression suite
- Designed a hybrid semantic search engine (local embeddings + Postgres full-text, fused via Reciprocal Rank Fusion) over ~27K chunked research documents, raising recall@5 from 0.917 to 1.000 and MRR from 0.889 to 0.958 on a scored benchmark
- Deployed local GPU speech-to-text (faster-whisper, int8) to transcribe podcast sources with no published transcript at ~75x realtime, cutting a ~2,200-episode backlog to 40-50 compute-hours via resumable runs

Skills

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- **Languages:** C#, Java, Python (pandas, NumPy, scikit-learn), SQL, JavaScript
 - **Backend & Infra:** ASP.NET Core, REST API design, Redis, Kubernetes, Docker, Azure (Entra, Key Vault), PKI/Keyfactor, PostgreSQL, SQLite, SQL schema design, FastAPI, job orchestration
 - **CI/CD & DevOps:** GitHub Actions, Jenkins, ArgoCD, Artifactory, GitOps
 - **Data & ML:** Hybrid/RAG search, local embeddings (Ollama), GPU inference (faster-whisper, CTranslate2), XBRL parsing
 - **Web:** React, Node.js, HTML/CSS
 - **Observability:** Datadog, Prometheus, OpenTelemetry